North South University

HW : 09

SUBMITTED BY :

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HW-09

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Ans. to the question no-01

(₹) List based on arrays:

(a) contains:

Time complexity: O(n)

(b) Insert:

Time complexity: O(1)

(₹) Linked list:

(a) contains:

Time complexity: O(n)

(b) Insert:

Time complexity:

with tail: O(1)

without tail: O(n)

④ Trees:

@ contain:

Time complexity: O(logn)

ⓑ Insert:

Time complexity: O(logn)

⑤ Hash table:

ⓐ contain:

Time complexity: O(n)

ⓑ Insert:

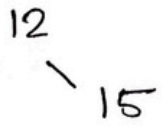
Time complexity: O(1)

Ans. to the question-02

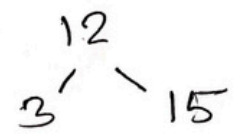
(a) Given numbers: 12, 15, 3, 35, ~~31~~ 21, 42, 14

insert 12:

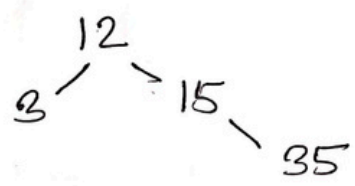
insert 15:



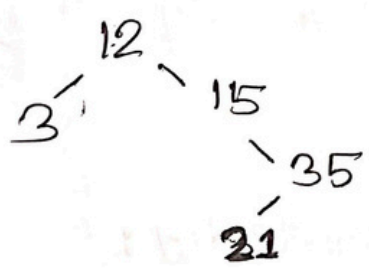
insert 3:



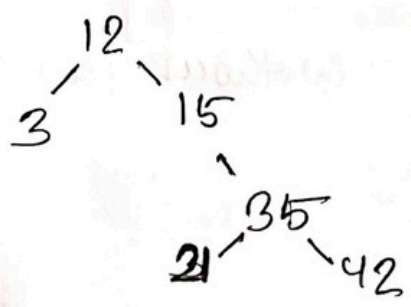
insert 35:



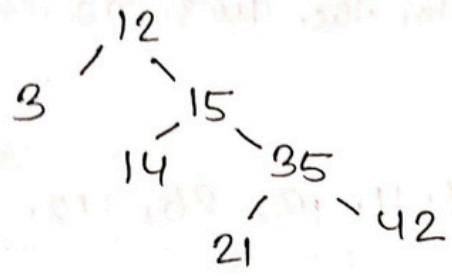
insert 21:



insert 42:



insert 14:



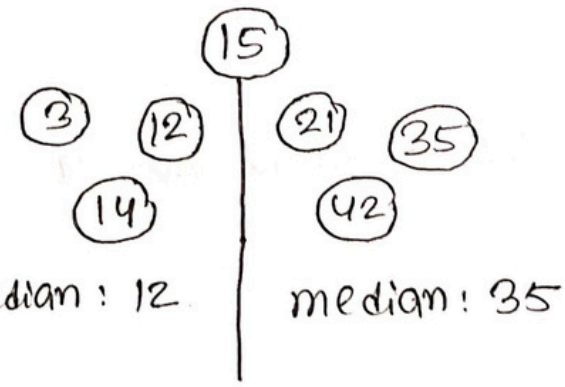
(b) Given numbers: 12, 15, 3, 35, 21, 42, 14

in sorted order : 3, 12, 14, 15, 21, 35, 42

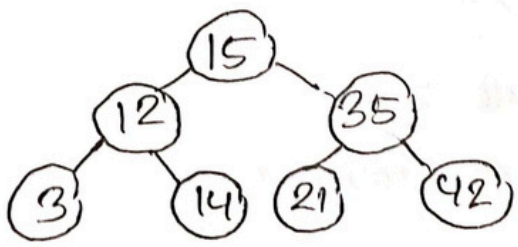
median : 15 so root : 15.

now, 3, 12, 14 at the left of 15 and 21, 35, 42 are at the right of 15

Step : 01



Step : 02



This is the balanced binary search tree.

Ans. to the ques. no-03

Pre order: 30, 24, 11, 13, 26, 40, 58, 48.

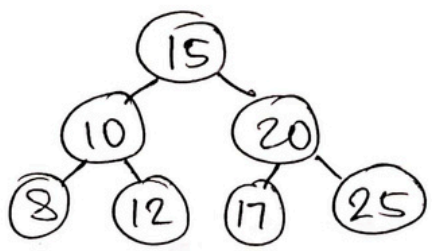
Post order: 13, 11, 26, 24, 48, 58, 40, 30

in order: 11, 13, 24, 26, 30, 40, 48, 58

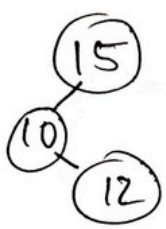
for a BST the inorder traversal always prints the keys in sorted order because the left subtree contains smaller element and right one contains larger ~~component~~ elements

(4)

Given tree:



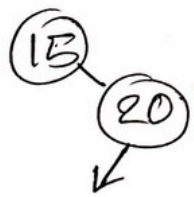
① find (12, R)



Nodes visited: 15, 10, 12

Find output: 12

② find (18, R)



Nodes visited: 15, 20

key not in tree

③ Time complexity in coonst case $O(\log n)$.

Modified algorithm :

```
Function find (k, R):
IF R is ~~not~~ null :
    return "key not found"
if R.key == k:
    return R
else if R.key > k:
    return find (k, R.Left)
else
    return find (k, R.Right)
```

Ans. to the QUESTION - ~~06~~ 05